Training Day 6

1. Topics Covered and Brief Explanation

1. Introduction to NumPy

NumPy (Numerical Python) is a popular Python library used for numerical computing. It provides support for working with multidimensional arrays and performing mathematical operations efficiently. It is widely used in Data Science and Machine Learning.

2. Features of NumPy

- Supports multidimensional arrays.
- Performs mathematical calculations efficiently.
- Provides functions for statistical operations.
- Supports matrix operations.
- Works with other Python libraries such as Pandas and Scikit-learn.

3. Array Creation

Learned how to create one-dimensional and two-dimensional arrays using the array() function.

Example:

```
import numpy as np

arr = np.array([10, 20, 30, 40])

print(arr)

matrix = np.array([[1, 2], [3, 4]])

print(matrix)
```

4. Array Indexing and Slicing

Indexing is used to access individual elements, whereas slicing is used to access a specific portion of an array.

Example:

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50])

print(arr[0])

print(arr[1:4])
```

5. Mathematical Operations

Studied basic mathematical operations such as addition, subtraction, multiplication, division and finding the mean of array elements.

Example:

```
import numpy as np
a = np.array([10, 20, 30])
b = np.array([1, 2, 3])
print(a + b)
print(a * b)
print(np.mean(a))
print(np.max(a))
```

2. Tools and Technologies Used

Tool	Purpose
Python 3	Programming language for numerical operations.
NumPy	Creating arrays and performing mathematical calculations.
Google Colab	Writing and executing NumPy programs.
Google Chrome	Accessing NumPy documentation and tutorials.

3. Practical Learning / Activities

- Created one-dimensional and two-dimensional NumPy arrays.
- Practised accessing array elements using indexing.
- Performed slicing operations on arrays.
- Executed mathematical and statistical operations.
- Explored how NumPy can be used to handle numerical data in machine learning.

4. Learning Outcomes

- Understood the purpose and importance of NumPy.
- Learned to create and manipulate arrays.
- Gained knowledge of array indexing and slicing.
- Practised mathematical and statistical operations.
- Developed a basic understanding of numerical data handling for AI and ML applications.